	SVKM's NMIMS **Mukesh Patel School of Technology Management & Engineering /** **School of Technology Management & Engineering**	
Program: B. Tech CE	**Lab and Workbook**	**Academic Year- 2026-27**
Year:-Third	**Subject:-** Drone Technology	**Semester: - Fifth**

Experiment: 8

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Aim:

- To study how to program Pixhawk Flight controller.
- To learn set up for R Channels.
- To program Pixhawk Flight controller connected drone.

Prerequisites: Basic knowledge of UAVs, Experience with radio transmitters and receivers, Basic programming knowledge

Learning outcomes:

After completing the lab, students will be able to:

1. **Understand the architecture and functionality of the Pixhawk Flight Controller.**
2. **Configure and program the Pixhawk using Mission Planner or QGroundControl.**
3. **Set up and calibrate radio channels (R Channels) for communication between transmitter and receiver.**
4. **Successfully program a Pixhawk-connected drone for autonomous or manual flight.**

Material Required

1. Flight Controller (PX4 or Ardupilot firmware)
2. Computer with Mission Planner or QGroundControl installed
3. RC Transmitter and Receiver (with at least 6 channels)
4. Drone frame and motors
5. Power module
6. USB cable for programming
7. Battery and other drone accessories

Theory:

Pixhawk Flight Controller: Pixhawk is an open-hardware autopilot that runs ArduPilot or PX4 firmware. It reads sensors, estimates the vehicle state, runs the control loops and drives the motors/servos; it can also fly pre-programmed autonomous missions.

Block	Details (Pixhawk 1)
Processor	32-bit STM32F427 Cortex-M4F @ 168 MHz, 256 KB RAM, 2 MB flash
Failsafe co-processor	STM32F100 (32-bit) – keeps manual override if main processor fails
Sensors	MPU6000 accelerometer/gyro, L3GD20 gyro, LSM303D accel/magnetometer, MS5611 barometer
Outputs	8 MAIN OUT + 6 AUX PWM outputs (motors/servos)
Inputs / ports	RC IN (PPM/SBUS), POWER, GPS, TELEM1/2 (radio), USB, CAN, I2C,

	ADC
Firmware	ArduPilot (Copter/Plane/Rover) or PX4; flashed using Mission Planner or QGroundControl

R (Radio) Channels: each stick or switch on the transmitter is one channel, sent to the receiver as a pulse width between 1000 µs (minimum) and 2000 µs (maximum), with 1500 µs at centre. The standard ArduCopter mapping is CH1 = Roll, CH2 = Pitch, CH3 = Throttle, CH4 = Yaw, CH5 = Flight-mode switch, CH6–CH8 = auxiliary functions. Radio calibration teaches the autopilot the minimum, maximum and trim values of each channel.

Ground control software: Mission Planner (ArduPilot, Windows) or QGroundControl (PX4/ArduPilot, multi-platform) is used for firmware loading, sensor calibration, radio setup, parameter tuning and mission planning.

Procedure:

1. Setup Pixhawk Flight Controller:

1. **Install Ground Control Software:**
   - Download and install **Mission Planner** (for ArduPilot firmware) or **QGroundControl** (for PX4 firmware) on your computer.
   - Connect the **Pixhawk flight controller** to your computer using a USB cable.
2. **Load Firmware:**
   - Open the ground control software and connect the Pixhawk.
   - Select the appropriate firmware for your drone. Choose between **ArduPilot** or **PX4**.
   - Flash the firmware onto the Pixhawk.
3. **Calibrate Sensors:**
   - After flashing the firmware, follow the on-screen instructions to calibrate the following sensors:
     - Accelerometer
     - Gyroscope
     - Compass
     - RC Transmitter

2. Setting up R Channels:

1. **RC Transmitter Setup:**
   - Connect the RC receiver to the Pixhawk using the provided connections (PPM/SBUS).
   - Turn on your RC transmitter and ensure the channels are functioning correctly.
2. **Mapping R Channels:**
   - In **Mission Planner/QGroundControl**, go to the "Radio Calibration" tab.
   - Move the sticks on the RC transmitter and observe the corresponding channel movements on the screen.
   - Assign each channel to its respective function (e.g., throttle, pitch, roll, yaw).
3. **Setting Mode Switch:**
   - Assign one of the RC channels (usually Channel 5) to the **flight mode switch**.
   - Configure different flight modes like **Stabilize**, **Altitude Hold**, **Loiter**, and **Auto**.

3. Programming the Pixhawk Connected Drone:

1. **Set Parameters for Flight:**
   - Go to the "Full Parameter List" or "Full Parameter Tree" in the ground control software.
   - Set basic parameters for drone control, such as:
     - **Throttle mid-point**
     - **Roll, pitch, and yaw sensitivity**
     - **Failsafe settings**
2. **Assign Autonomous Missions (Optional):**

- In **Auto mode**, you can set waypoints for the drone to follow.
- Use the ground control software to define a mission by clicking on the map to add waypoints.
- Upload the mission to the Pixhawk.

3. **Test the Drone:**
   - After completing the setup, place the drone in a safe area.
   - Arm the drone by using the arming switch or by moving the throttle stick on your RC transmitter.
   - Test the basic manual flight modes first (e.g., Stabilize).
   - Once comfortable, test the autonomous flight mode (if programmed).

PART B

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Date of Experiment: 15/10/24	**Date of Submission:** 15/10/24

Task to be performed

1. Pixhawk Setup & Connection
2. Radio Channels (R Channels) Setup & Calibration
3. Programming the Pixhawk Flight Controller
4. Motor & ESC Testing

Task 1: Pixhawk Setup & Connection

- Installed Mission Planner, connected the Pixhawk through USB and selected the correct COM port (baud 115200) → Connect.
- Initial Setup → Install Firmware: selected ArduCopter and the frame type (Quad X) and flashed the firmware.
- Calibrated the accelerometer (6 orientations), compass (rotate in all axes) and checked the gyroscope/level calibration.

Task 2: Radio Channels (R Channels) Setup & Calibration

- Connected the FS-iA6B receiver to the Pixhawk RC IN port (PPM) or to a serial port (i-BUS) and powered the transmitter.
- Initial Setup → Mandatory Hardware → Radio Calibration → Calibrate Radio: moved all sticks and switches to their extremes and clicked Done; the green bars showed minimum/maximum for each channel.

- Verified the mapping – Roll (CH1), Pitch (CH2), Throttle (CH3), Yaw (CH4) – and reversed a channel if the bar moved opposite to the stick.
- Flight Modes: set CH5 as the mode switch and assigned Stabilize, Alt Hold, Loiter, Auto, RTL and Land to the six PWM ranges (FLTMODE1–FLTMODE6).

Channel	Function	PWM range (µs)
CH1	Roll	1000 – 2000
CH2	Pitch	1000 – 2000
CH3	Throttle	1000 – 2000
CH4	Yaw	1000 – 2000
CH5	Flight mode switch	1000 – 2000 (6 mode bands)
CH6–8	Auxiliary (tuning, camera, etc.)	1000 – 2000

Task 3: Programming the Pixhawk Flight Controller

Parameter	Purpose / value
FRAME_CLASS / FRAME_TYPE	Frame selection – Quad, X
THR_MID / MOT_THST_HOVER	Throttle mid-point/hover throttle (≈ 500 or 0.3–0.4)
ATC_ANG_RLL_P, ATC_ANG_PIT_P, ATC_RAT_*	Roll/pitch/yaw sensitivity and PID gains
FS_THR_ENABLE	Radio failsafe action (e.g. 1 = RTL)
BATT_MONITOR, BATT_FS_LOW_ACT	Battery monitor and low-battery failsafe
ARMING_CHECK	Pre-arm checks enabled (1 = all)

- Opened Config/Tuning → Full Parameter List, searched for the parameters above, changed values and clicked Write Params.
- Created a simple mission in Flight Plan (Takeoff → waypoints → RTL) and uploaded it with Write WPs for AUTO mode.

Task 4: Motor & ESC Testing

- Removed all propellers. Performed ESC calibration (all-at-once method) so that every ESC learns the same minimum and maximum throttle.
- Initial Setup → Optional Hardware → Motor Test: tested motors A, B, C, D at about 5–10 % throttle for a few seconds each and checked rotation direction and order (A = Motor 1 … D = Motor 4). Swapped two motor wires where the direction was wrong.
- Armed in Stabilize mode (propellers still off), raised the throttle slowly and confirmed all four motors respond equally; then disarmed.

Safety: always remove propellers during configuration and motor tests.

Observations and Learning:

- The Pixhawk is connected to the PC by USB; Mission Planner flashes the firmware and calibrates the accelerometer, compass and radio.
- Radio calibration shows each channel moving between 1000 and 2000 µs; mapping CH1–CH4 and a mode switch on CH5 is required for correct control and for switching between Stabilize, Alt Hold, Loiter and Auto.
- Parameters such as THR_MID, failsafe and sensitivity settings change how the drone responds; failsafes must be enabled before flight.
- Motor test confirmed correct motor order/direction and that all ESCs are calibrated equally.
- Manual mode was tested first and autonomous mission only after manual flight was stable.

Conclusion:

The architecture and functionality of the Pixhawk flight controller were understood. The Pixhawk was configured and programmed with Mission Planner (firmware, sensor calibration, parameters), the R channels of the FlySky transmitter and receiver were set up and calibrated including the flight-mode switch, and a Pixhawk-connected drone was prepared for manual and autonomous flight after motor and ESC testing. All four learning outcomes were attained.